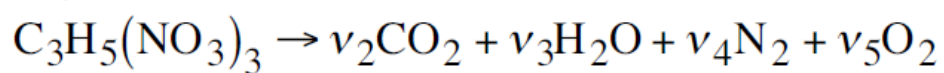


P1.2

From element balances on N, C, H, and O, we write 4 equations:

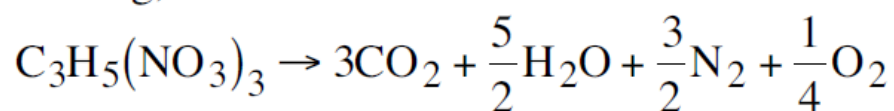
$$3 = 2\nu_4$$

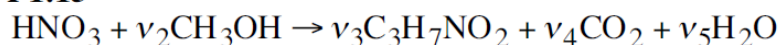
$$3 = \nu_2$$

$$5 = 2\nu_3$$

$$9 = 2\nu_2 + \nu_3 + 2\nu_5$$

Solving, we find



P1.15

The element balance equations for N, C, H and O are

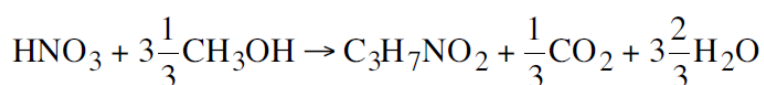
$$1 = \nu_3$$

$$\nu_2 = 3\nu_3 + \nu_4$$

$$1 + 4\nu_2 = 7\nu_3 + 2\nu_5$$

$$3 + \nu_2 = 2\nu_3 + 2\nu_4 + \nu_5$$

This is a set of 4 equations in 4 unknowns that we solve by substitution and elimination to find the balanced reaction:



We want to react (54-10 mg/L) x 10 L of nitric acid, or 0.44 g. The molar mass of HNO_3 is 63 g/gmol, while that of CH_3OH is 32 g/gmol. Therefore:

$$0.44 \text{ g HNO}_3 \times \frac{\text{gmol HNO}_3}{63 \text{ g HNO}_3} \times \frac{3\frac{1}{3} \text{ gmol CH}_3\text{OH}}{\text{gmol HNO}_3} \times \frac{32 \text{ g CH}_3\text{OH}}{\text{gmol CH}_3\text{OH}} = 0.75 \text{ g CH}_3\text{OH}$$

P1.17

$$\text{Ethanol: } \frac{6 \text{ gmol H}}{\text{gmol C}_2\text{H}_5\text{OH}} \times \frac{\text{gmol C}_2\text{H}_5\text{OH}}{46 \text{ g C}_2\text{H}_5\text{OH}} \times \frac{1 \text{ g H}}{\text{gmol H}} \times 100\% = 13\text{wt}\% \text{ H}$$

$$\text{Water: } \frac{2 \text{ gmol H}}{\text{gmol H}_2\text{O}} \times \frac{\text{gmol H}_2\text{O}}{18 \text{ g H}_2\text{O}} \times \frac{1 \text{ g H}}{\text{gmol H}} \times 100\% = 11\text{wt}\% \text{ H}$$

$$\text{Glucose: } \frac{12 \text{ gmol H}}{\text{gmol C}_6\text{H}_{12}\text{O}_6} \times \frac{\text{gmol C}_6\text{H}_{12}\text{O}_6}{180 \text{ g C}_6\text{H}_{12}\text{O}_6} \times \frac{1 \text{ g H}}{\text{gmol H}} \times 100\% = 6.7\text{wt}\% \text{ H}$$

$$\text{Methane: } \frac{4 \text{ gmol H}}{\text{gmol CH}_4} \times \frac{\text{gmol CH}_4}{16 \text{ g CH}_4} \times \frac{1 \text{ g H}}{\text{gmol H}} \times 100\% = 25\text{wt}\% \text{ H}$$

It does seem hard to believe that they achieved 50 wt% H.